RP2040 chord style VCO by Hagiwo
SMD version by BleepSound

replace output with a DAC ?
See DAC8164ICPW for example.
Would force a code modification
https://deepbluembedded.com/digital-to-analog-converter-dac-waveform-generation-with-mcu/

H3 H4 MountingHole_Pad
H1 H2 SpacerHoles

U2 Seed_XIAO_RP2040
V/oct 1 D0 D7 8 SW_1
Tune 2 D1 D8 9 Out
Inv 3 D2 D9 10
Chord 4 D3 D10 11
Push_SW 5 D4
SW_0 6 D5
D6
3V3 12 +3V3
5V 14 +5V
GND 13 GND

TUNE1 10k TunePot R5 1k Tune C4 100nF
CHORD1 10k ChordPot R6 1k Chord C7 100nF
C5 100nF C6 10uF

Out R2 1k C2 10nF R3 1k C3 10nF
U1A TL074
R1 6.8k
R4 330R C1 10nF OutJack
J1 Out

SW1 SW_MEC_LOW_PROF_LED
R17 10k Push_SW
+12V R7 330R L1 L2 GND

SW2 SW_SPDT SW_0 SW_1

J2 GND InvPot TunePot ChordPot +5V +3V3
J3 GND InvPot TunePot ChordPot +5V +3V3
J4 OutJack InvJack V/octJack GND SW_0 Push_SW SW_1 +12V
J5 OutJack InvJack V/octJack GND SW_0 Push_SW SW_1 +12V

J6 V/oct S SN TN GND V/octJack
R9 220k
D1 BAT43 +3V3
R11 330k D2 BAT43
V/oct C8 10nF

INV1 10k InvPot R8 10k
U1B TL074
R10 10k

J7 Inv CV S SN TN GND InvJack
R18 10k
R12 10k
U1C TL074
R14 10k
R13 10k
U1D TL074
R15 22k
R16 33k D4 BAT43
D3 BAT43 Inv C9 10nF

Power Input J8
10 9 8 7 6 5 4 3 2 1
C10 10uF C11 100nF +12V
GND
C15 10uF C16 100nF -12V

U1E TL074 +12V -12V

U3 AP2210K-5.0TRG1
VIN VOUT +5V
EN
GND BYP
C13 1uF C12 2.2uF C14 100pF

Bleep Sound
Sheet: /
File: RP2040-VCO.kicad_sch
Title: RP2040 VCO
Size: A4 Date: 2025-10-29 Rev: 1.5
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